

- 2 (a) State the *first law of thermodynamics*.

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..... [2]

The increase in internal energy of a closed system is equal to the sum of B2
heat supplied to the system and the work done on the system.

B1 – Correct quantities referred to

B1 – Directions clearly and correctly stated. Use of “sum of” or equivalent.

- (b) A fixed mass of monatomic ideal gas undergoes a cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ as shown in Fig. 2.1.

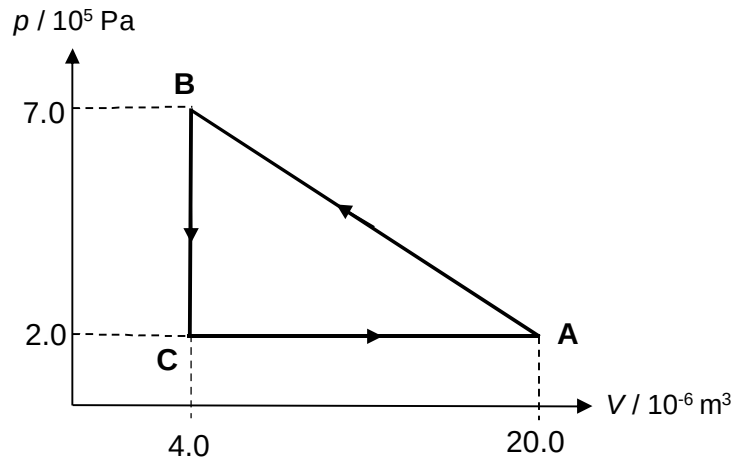


Fig. 2.1

- (i) Explain why the change in internal energy of the ideal gas during a complete cycle $A \rightarrow B \rightarrow C \rightarrow A$ is zero.

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..... [2]

L2 In a complete cycle, the (p , V) state of the gas returns to the original state. **B1**
Therefore, using the equation of state of an ideal gas, the temperature of the gas also returns to the original temperature.

As the internal energy of an ideal gas is proportional to the thermodynamic temperature of the gas, there is no net change in the internal energy in a complete cycle. **B1**

(ii) Calculate the work done on the gas during the process from A to B.

L1 work done on gas = J [2]
= area under the graph
$$= \frac{1}{2}[(7.0+2.0) \times 10^5][(20.0-4.0) \times 10^{-6}]$$
 C1
$$= 7.2 \text{ J}$$
 A1

(iii) Calculate the change in internal energy of the gas when it undergoes the process from A to B.

L2 Change in internal energy = J [2]
For a monoatomic ideal gas,
$$\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \Delta(pV)$$

M1

change in internal energy

A1

$$= \frac{3}{2} [(7.0 \times 10^5)(4.0 \times 10^{-6}) - (2.0 \times 10^5)(20.0 \times 10^{-6})]$$
$$= -1.8 \text{ J}$$

(iv) Calculate the heat supplied to the ideal gas during the process from A to B.

heat supplied = J [2]

L1 Using 1st law of thermodynamics,

$$\Delta U = Q + W$$

$$Q = -1.8 - (7.2)$$

$$= -9.0 \text{ J} \quad (\text{No Negative, No marks})$$

C1

A1

(v) In a car engine, the expanding combustion gases push the piston which causes the car to move.

Suggest, with a reason, whether the cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ could be the cycle in the piston engine of a car.

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L3 Not possible, because for a car to move, there must be a positive net work done by the gas per cycle in the piston engine. (i.e. net work done on gas is negative.)

Whereas the net work done by the gas in the cycle $A \rightarrow B \rightarrow C \rightarrow A$ is negative.

[1]

B1